[Assignment-1]

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Answer to the question no 1

a

Heuristic is admissible, if $h(n) \leq h^*(n)$ is true for every node n. where, $h^*(n)$ is the true cost to reach the goal from n.

States, n	h(n)		$h^*(n)$	Status
S	2	<	4	True
A	4	<	9	True
B	1	<	3	True
C	2	<	3	True
D	1	<	2	True
E	3	<	5	True
G	0	$\leq$	0	True

Here, the heuristic is admissible.

b

For GBFS,

we know that, $f(n) = \cancel{g(n)+}\ h(n)$

Here, we maintain a priority queue. which is called fringe.

$\boxed{S^2}$

S^2 $\boxed{A^4 \ B^1 \ C^2}$

B^1 $\boxed{A^4 \ C^2 \ G^0}$

G^0 $\boxed{A^4 \quad C^2}$

Path : S → B → G

For A* search,

we know that, $f(n) = h(n) + g(n)$

we will use priority queue (fringe) again.

$\boxed{S^2}$

S^2 $\boxed{A^5 \quad B^2 \quad C^4}$

B^2 $\boxed{A^5 \quad C^4 \quad G^4}$

C^4 $\boxed{A^5 \quad G^4 \quad D^4}$

G^4 $\boxed{A^5 \quad D^4}$

state, n	h(n)	g(n)	h(n)+g(n)	Parent
S	2	0	2	
A	4	1	5	S
B	1	1	2	S
C	2	2	4	S
D	1	3	4	C
E	3	5	8	A
G	0	4	4	B

Path : S → B → G

a

A heuristic is consistent if for every node n, every successor n' of n generated by an action a, follows the triangular inequality.

$$h(n) \leq c(n, a, n') + h(n')$$

If h is consistent,

$$f(n') = g(n') + h(n')$$

For,

Node S: $h(S) = 2 \leq 3$	$c(S,A) + h(A) = 1 + 2 = 3$
$h(S) = 2 \leq 2$	$c(S,B) + h(B) = 1 + 1 = 2$
$h(S) = 2 \leq 4$	$c(S,C) + h(C) = 2 + 2 = 4$
Node A: $h(A) = 4 \leq 7$	$c(A,E) + h(E) = 4 + 3 = 7$
Node B: $h(B) = 1 \leq 3$	$c(B,G) + h(G) = 3 + 0 = 3$
Node C: $h(C) = 2 \leq 2$	$c(C,D) + h(D) = 1 + 1 = 2$
Node D: $h(D) = 1 \leq 2$	$c(D,G) + h(G) = 2 + 0 = 2$

Node E : $h(E) = 3 \le 3$	$c(E,B) + h(B)$ $= 2 + 1 = 3$

~~Node G : $h(G) = 0 \le 0$~~

So, the heuristic is consistent.

d

GBFS problems solved by A^* search :

1) **Non-optimality :** GBFS may follow a deceptive heuristic path and miss the optimal route, whereas A^* with an admissible heuristic guarantees optimality.

2) **Completeness / Infinity Loops :** GBFS can loop in infinite trees if heuristic misleads. A^* avoids revisiting cheaper paths.

3) **Path cost ignoring :** GBFS ignores path cost $g(n)$, risking long detours. A^* consider both $g(n)$ and $h(n)$; $h(n)$ = heuristic value
$g(n)$ = actual path cost.

Answer to the question no 2

a

5x5 grid ⇒

	0	1	2	3	4
0	4	3	2	1	$\boxed{0}$
1	5	4	3	2	1
2	6	5	4	3	2
3	7	6	5	4	3
4	$\boxed{8}$	7	6	5	4

[Manhattan Distance] ($h_1(n)$)

For $(1,0)$, $h_1 = 5$

and, Actual path cost $= \sqrt{2} + 1 + 1 + 1$

$= 3 + \sqrt{2}$

$= 4.41$

so, $h_1(n) \nless g(n)$

For $(2,0)$, $h_1 = 6$

and, Actual path cost $= \sqrt{2} + \sqrt{2} + 1 + 1$

$= 4.83$

so, $h_1(n) \nless g(n)$

so, $h_1(n)$ is not admissible.

For, $h_2(n)$ [Euclidean Distance]

when,

$$(1,0),\ h_2 = \sqrt{(0-1)^2+(0-4)^2} = \sqrt{1^2+4^2} = \sqrt{17} = 4.12$$

and, actual path cost $= \sqrt{2}+1+1+1$

$= 4.41$

So

For,

$$(2,0),\ h_2 = \sqrt{(0-2)^2+(0-4)^2} = \sqrt{20} = 4.47$$

and actual path cost $= \sqrt{2}+\sqrt{2}+1+1$

$= 4.83$

so, $h_2(n) \leq g(n)$

For,

$$(1,0),\ h_2 = \sqrt{(0-1)^2+(0-4)^2} = \sqrt{17} = 4.12$$

and actual path cost $= \sqrt{2}+1+1+1$

$= 4.41$

so, $h_2(n) \leq g(n)$.

So, $h_2(n)$ is admissible.

b

Here,

$$h_1(n)' = 1.2 \times h_1(n)$$

Since h_1 already overestimated some true costs, scaling will only increase overestimation. Thus h_1' is not admissible.

without admissibility, A* search with h_1' can lose optimality.

c

When multiple admissible heuristics exist, we will take the dominant one (ex- the one with higher values without overestimating), which yields fewer expansions and faster search.

$$h(n) = \max\{h_1(n), h_2(n)\}$$